

Xuanzhi (玄 智)

A GraphRAG-powered cognition layer that augments curious non-experts into rigorous PhD-like researchers.

Pathway: 1 — Your Own 10x Idea

Program: Management & Xponential Technology (MXT) — IE University/IE Business School/IEX Research Xcelerator

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Executive Abstract

Xuanzhi is a GraphRAG-native research cognition platform designed to close the gap between the curious non-expert and the PhD-trained researcher. Where today's AI research tools optimise for retrieval and summarisation, Xuanzhi scaffolds the harder, scarcer skills that define rigorous inquiry: abstraction, synthesis, hypothesis generation, and research judgement under uncertainty. The product is built on a graph-based knowledge architecture (Microsoft's GraphRAG), frontier reasoning models, and software-encoded research methodology. A functional v0 prototype: `research_automator`, developed during an Operating Systems & Parallel Computing course, already validates the ingestion and knowledge-graph pipeline end-to-end at near-zero cost. The 10x leap is the cognitive scaffolding layer we propose to build on top of it. The goal is not a faster search engine. It is what calculators did for arithmetic: not to replace thought, but to dramatically extend who can do high-level intellectual work well.

1. Company and Context Overview

"玄之又玄，众妙之门"

Global scientific output has roughly doubled in under a decade, rising from about 1.71 million indexed research studies in 2015 to approximately 2.53 million in 2024, with total article output (all document types) approaching 3.3 million per year and growing at roughly 5-7% annually. The race to keep pace with this volume has spawned a generation of AI-native research tools, Elicit, Consensus, Perplexity, Scite, Undermind, SciSpace, NotebookLM, and native research modes inside Anthropic's and OpenAI's flagship chat products, that compete primarily on the axes of speed, breadth, and summary quality.

The dominant technical paradigm across this competitive set is Retrieval-Augmented Generation (RAG): a user asks, the system retrieves relevant document chunks, and the LLM writes an answer. This architecture handles "what does this document say?" well, but it breaks down on the questions that define real research, questions about assumptions, contradictions, schools of thought, and the general principles that can be abstracted across cases. Microsoft Research has quietly opened a new front here with Project GraphRAG, which combines text extraction, network analysis, and LLM summarisation to support multi-hop reasoning over LLM-derived knowledge graphs, and whose newer LazyGraphRAG and DRIFT Search variants

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are now production-viable on Azure. To date, however, GraphRAG is being deployed primarily as enterprise infrastructure (for example, inside Microsoft Discovery, an agentic platform for scientific R&D on Azure) rather than as a consumer-facing product category.

Xuanzhi's strategic positioning sits deliberately off-axis from the current competitive set. We are not the next summariser. We are the research cognition layer: the only category in which the customer value proposition is "you will think better, not just search faster." GraphRAG is the enabling technology; cognitive scaffolding is the product.

2. Problem Statement

There is a real and widening gap between the ordinary curious person and a PhD-trained researcher. Current AI tools have compressed the retrieval gap impressively, but the cognition gap is largely untouched. Five capabilities define research expertise, and none are adequately scaffolded by today's products: abstraction from particulars to generalisable patterns; problem formulation and hypothesis generation; multi-source conceptual synthesis; research judgement under uncertainty; and methodological self-criticism. These capabilities are learned through years of structured mentorship, not through a better search box.

The scale of the underserved population is enormous. UNESCO estimates over 8.8 million researchers globally. Universities enrol more than 250 million students at any given time, a large fraction of whom are asked to produce research-quality work without formal research training. Beyond academia, journalists, policy analysts, founders doing market and technical discovery, consultants, autodidactic professionals, and a growing class of AI-assisted generalists are all routinely forced to do PhD-shaped thinking with shopkeeper-grade tools. The volume problem compounds the skill problem: researchers already spend an estimated 130 million hours per year on peer review alone, and the exponential growth of the literature means even trained experts cannot keep pace with conventional reading.

The diagnosis of current tools is specific. Chatbots answer questions but do not structure an inquiry. Summarisers compress text but do not surface contradictions. RAG pipelines retrieve passages but do not map the conceptual relationships between them. Note-taking systems such as Obsidian, Roam, and Logseq allow users to build knowledge graphs by hand but impose the entire cognitive labour of graph construction on the human. None of these tools actively teaches abstraction, critiques under-determined inferences, or generates counter-hypotheses.

Direct prototype-based validation. Our predecessor project, *research_automator*, was built as a working proof that multi-source paper ingestion, LLM summarisation, and concept-linked knowledge-graph construction can be pipelined end-to-end at effectively zero marginal cost, roughly \$0.00002 per paper using Claude Haiku. The prototype achieves a ~99% paper capture rate across Google Scholar, Semantic Scholar, and ArXiv, exporting concept-linked Markdown into an Obsidian vault that the user then navigates as a knowledge graph. Running and stress-testing the prototype over several weeks surfaced precisely the ceiling of current approaches: it automates discovery and lightweight concept tagging beautifully, but then leaves the user alone in front of a steadily growing graph, with no coach to help abstract, no adversarial partner to stress-test hypotheses, and no scaffold for synthesis. That ceiling is where the 10x opportunity begins.

3. Proposed 10x Exponential Solution

Xuanzhi is an AI-native research environment that turns vague curiosity into defensible insight by scaffolding the cognitive operations of rigorous inquiry. The product stands on three deep-technology pillars, each on its own insufficient, and together architecturally novel.

GraphRAG as a cognitive substrate. Rather than embedding documents into flat vector stores, Xuanzhi uses LLM-derived knowledge graphs, built on Microsoft's open-source GraphRAG library and its LazyGraphRAG / DRIFT Search extensions, to represent concepts, claims, authors, methods, findings, and disagreements as interconnected entities. This is what enables multi-hop reasoning ("what assumptions sit underneath this debate?"), cross-source synthesis ("what schools of thought exist?"), and conceptual navigation ("where are the contradictions?"). These operations are architecturally impossible for chunk-based RAG.

Reasoning-model-orchestrated agentic workflows. Xuanzhi uses frontier reasoning models (Claude Opus, GPT-style reasoning variants, and open reasoning models) as research supervisors that plan multi-step inquiries, call the graph, critique their own outputs, and iterate. We adopt and extend Microsoft's own Claimify (claim extraction) and VeriTrail (hallucination detection and provenance tracing) techniques so that every synthesis step is traceable back to graph evidence.

Structured cognitive scaffolds: software-encoded research methodology. Xuanzhi exposes seven research primitives as first-class product surfaces: a Question Refiner (turns vague prompts into researchable sub-questions and conceptual dimensions); a Concept Graph Workspace (the user's visible thinking map, auto-built and hand-editable); an Abstraction Coach (asks what pattern repeats, what unifies these cases, what mechanism is at work); a Hypothesis Generator (proposes hypotheses and counter-hypotheses along with the evidence each would require); a Source-Grounded Synthesis Engine (produces argument maps, evidence tables, and competing-theory comparisons anchored in graph relationships, not just retrieved passages); Guided Research Modes (different tuning for beginner explorers, serious autodidacts, undergraduate thesis writers, policy researchers, journalists, and founders); and a Mentor Feedback Loop (a supervisor-style critic that tells users, "your question is too broad," "you are conflating two concepts," "this inference is under-supported," "this is descriptive, not explanatory").

This is a 10x leap rather than an incremental improvement for three reasons. First, it is a category move, not a feature race. Competitors optimise retrieval speed; Xuanzhi expands who can do rigorous research work at all. Second, it compounds through use; the user becomes measurably better at abstraction and synthesis, not just the outputs. That is the signature of a cognitive-augmentation product rather than a productivity tool. Third, it exploits a narrow technological window: GraphRAG only became practical at commodity cost in the last eighteen months, frontier reasoning models only in the last twelve. The combination unlocks a product category that was architecturally infeasible in 2023. The closest historical analogue is what spreadsheets did for financial modelling, or what calculators did for arithmetic; the tool did not replace thought; it dramatically extended who could do high-level intellectual work well.

4. Implementation Strategy

The roadmap builds directly on demonstrated work and proceeds in four phases. The strategic design principle is capital efficiency: the hardest and most expensive infrastructure layer (ingestion, graph construction, LLM orchestration) sits on commodity open-source GraphRAG plus pay-per-token reasoning models, which means Xuanzhi's defensibility does not come from infrastructure, but from cognitive scaffolds, workflow data, and distribution.

Phase	Timeline	Focus	Milestones & Partners
Phase 0	Complete	research_automator prototype: multi-source ingestion, LLM summarisation, concept-linked Markdown export to Obsidian.	Validated the full ingestion-to-graph pipeline at ~\$0.00002/paper and a 99% capture rate across Google Scholar, Semantic Scholar, and ArXiv.
Phase 1	Months 0–6	MVP: replace Obsidian-markdown graph with production GraphRAG (Microsoft OSS). Ship Question Refiner + Abstraction Coach as the differentiated MVP features on a web app.	5,000 active beta users; measurable improvement on user-defined research-quality rubrics. Partners: Anthropic (Claude reasoning), Microsoft (Azure + GraphRAG), OpenAlex, Semantic Scholar.
Phase 2	Months 6–18	Cognitive depth: Hypothesis Generator, Synthesis Engine, first three Guided Modes (UG thesis, policy, founder). Integrate Claimify + VeriTrail to harden against hallucination.	100,000 Pro subscribers; first institutional pilots with two universities and one think tank.
Phase 3	Months 18–36	Institutional layer: Mentor Feedback Loop with longitudinal skill tracking. Launch institutional tier (SSO, admin analytics, curriculum integration). Expose developer API.	1M total users; 500+ institutional seats; first data moat from accumulated reasoning-workflow telemetry.

Capability and partnership requirements are modest by design. The core technical team required for Phase 1 is small, four to six engineers covering GraphRAG infrastructure, LLM orchestration, and product/frontend, plus one research-methodology designer responsible for cognitive-scaffold design. Distribution in Phase 1 targets ambitious undergraduates and serious autodidacts through student-community partnerships and an Obsidian community plugin (a natural adjacency to the v0 prototype). Phase 2 adds institutional outreach via university research-methodology faculty, policy-research think tanks, and the YC founder community. Phase 3 scales through API partners who embed Xuanzhi's cognition primitives into vertical research agents for law, medicine, patent research, investigative journalism, and competitive intelligence.

5. Scalability & Business Logic

Xuanzhi scales along three independent vectors that compound.

Freemium consumer SaaS. A generous free tier supports occasional research sessions; Xuanzhi Pro at ~€20/month unlocks unlimited sessions, the full Guided Modes library, and the Mentor Feedback Loop; Xuanzhi Scholar, a reduced-price tier at ~€8/month, is gated behind verified university email domains. Unit economics are favourable and empirically grounded: the v0 prototype confirmed that ingestion and summarisation on Claude Haiku run at roughly \$0.00002 per paper. Even a heavy Pro user executing hundreds of reasoning-model operations per month costs approximately €2-5 in inference, implying gross margins of 75-90%.

Institutional B2B2C. Universities, think tanks, and corporate R&D labs buy seat-based licences with admin consoles, SSO, cohort analytics, and library-system integration. This tier drives higher ARPU (€200–500 per seat annually) and unlocks a distribution channel that is otherwise expensive to reach at the consumer level.

Developer platform. An API exposes Xuanzhi's cognition primitives, Question Refiner, Abstraction Coach, Hypothesis Generator, and Synthesis Engine, so that third parties can build vertical research agents. Platform economics are usage-based with 60-80% gross margins after inference cost.

Exponential scalability is structural rather than brute-force. Knowledge graphs exhibit power-law connectivity: each new user's queries and workflows enrich the shared concept graph with new relationships, which improves retrieval and reasoning for every subsequent user, a classic data network effect. Guided Research Modes are community-authored: domain experts can publish modes and earn revenue share, turning content acquisition into a marketplace rather than a cost centre. The Mentor Feedback Loop accumulates anonymised reasoning telemetry, a proprietary dataset that no incumbent can replicate from public sources and that compounds with every session. Together, these effects mean that Xuanzhi's marginal unit economics improve with scale rather than degrade with it.

6. Projected Impact Analysis

Quantitative impact. The addressable population includes roughly 8.8 million working researchers, over 250 million university students, several million journalists, policy analysts, and consultants, and an indeterminately large body of autodidacts and knowledge workers performing research-adjacent work, conservatively, a serviceable addressable population of 50–70 million users by Year 5. At a modest 5% adoption rate and an average of 2-3 hours of research work per user per week, Xuanzhi would augment approximately 300-500 million hours of structured inquiry annually. If even a conservative 20% productivity uplift is realised, well within ranges reported for AI-augmented knowledge work in early studies, this is equivalent to adding 30,000–50,000 full-time-equivalent researcher-years of cognitive output to the world each year.

Qualitative transformation. The larger impact is the democratisation of rigorous thinking. Today, high-quality research training is concentrated in a few dozen elite institutions worldwide. Xuanzhi substantially flattens that gradient: a motivated student at a regional university, a journalist at a local paper, or a founder in an emerging market gains access to the same cognitive scaffolding that PhD supervision provides at elite schools. The product upskills through use; users demonstrably improve at abstraction, question framing, and synthesis over time, so the tool leaves lasting capability behind even when users are not actively using it.

Organisational transformation for partner institutions. Universities adopting Xuanzhi at the curriculum level can redesign research-methodology courses around AI-native scaffolding, freeing faculty from repetitive supervisor overhead and concentrating their time on the irreducibly human parts of mentorship. For corporate R&D labs and think tanks, Xuanzhi compresses the time from hypothesis to defensible synthesis and raises the floor of junior researcher output.

Societal value and sustainability. Better-informed journalism and policy research feed into better public discourse; better-equipped founders make more realistic bets; better-trained students graduate with transferable cognitive skills. Xuanzhi aligns directly with UN SDG 4 (Quality Education), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 10 (Reduced Inequalities) through cognitive access rather than mere information access.

Global scalability. The product is digitally native, language-agnostic (frontier models now handle 50+ languages at research-adequate quality), and unconstrained by geography, physical capital, or licensing regimes. The marginal cost of launching in a new country is localisation and compliance; unit economics do not degrade with geography.

7. Risks & Mitigation

Exponential ambition requires responsible foresight. Xuanzhi faces five categories of risk, each with concrete, product-level mitigation strategies rather than disclaimers.

Risk category	Principal threat	Mitigation
Technological	Hallucination in multi-step synthesis; plausible claims not grounded in the source graph.	Architectural mitigation: every synthesis statement tethered to graph nodes via Claimify-style claim extraction and VeriTrail-style provenance tracing, exposed in the UI. Untraceable claims are flagged and visually degraded. LazyGraphRAG keeps compute costs in check.
Market competitive /	Commoditisation by OpenAI, Anthropic, Google, or Microsoft, adding a “research mode” to their flagship chat products.	Depth (cognition scaffolds are multi-year product builds, not a weekend of prompt engineering); institutional contracts that create real switching costs; category framing above the retrieval-and-summary fight the incumbents are likely to dominate.
Regulatory	EU AI Act, emerging US frameworks, and academic-integrity regulations.	Ship transparency and auditability as product features (every synthesis traceable, every Mentor critique loggable). Partner with anti-plagiarism providers rather than compete. Design for compliant academic use cases with clear AI-assistance disclosure.
Ethical	Cognition-augmentation product degrading into a cognition-replacement product — users outsourcing thinking rather than developing it.	Designed into the product: Mentor Feedback Loop requires user reasoning steps before granting synthesis outputs; progress measured in skill growth (user-authored abstractions, hypothesis quality, self-revision rates) rather than speed; Guided Modes demand more user cognition at higher skill levels, not less.
Business	Incumbent research-assistant startups (Elicit, Consensus, Undermind, Perplexity, SciSpace) are moving upmarket toward synthesis-style features.	Velocity on the category move rather than defensive feature parity. Cognition scaffolding is a 5-year build that competitors are organisationally unlikely to prioritise. Early institutional contracts and the developer API raise switching costs structurally.

Beyond the table, Xuanzhi commits to an annual transparency report covering model usage, failure modes, skill-growth measurements, and known limitations; and to an independent academic advisory board that reviews product decisions with pedagogical implications. These are not compliance tax lines, they are central to the brand promise that serious users and institutions will require from a product that explicitly claims to improve how people think.

Appendix

A1.v0 Prototype Architecture: [research_automator](#)

The v0 prototype is an OSPC-course project (Python 3 / Selenium / Claude Haiku) that validates the ingestion and graph-export pipeline on which Xuanzhi's GraphRAG layer will be built. The pipeline discovers academic papers across Google Scholar, Semantic Scholar, and ArXiv using a headless browser with human-like delays to prevent blocking; extracts titles, authors, and abstracts; calls the Anthropic API (Claude Haiku) to generate 2-3 sentence summaries and extract key concepts; and writes per-paper Markdown files into an Obsidian vault with wiki-style concept links. Concept files are auto-generated as hub pages, and Obsidian's native Graph View renders the result as a live, navigable knowledge graph. Observed performance: ~99% capture rate, 5-10 minute end-to-end run per batch, ~\$0.00002 per paper in API cost, and daily scheduled runs via cron. Source: github.com/devMYurge/research_automator.

A2.What [research_automator](#) proves and what it deliberately leaves to Xuanzhi

The prototype proves three things that materially de-risk Xuanzhi: (i) multi-source academic ingestion is robust and cheap at the individual-user scale; (ii) LLM-generated concept extraction is good enough to seed a usable knowledge graph; and (iii) a graph-shaped artefact genuinely changes the quality of user exploration relative to a flat reading list. It deliberately leaves four things undone, each of which is an Xuanzhi build: replacing the naïve concept-link graph with a GraphRAG-derived relational graph including entities, claims, methods, and disagreements; adding reasoning-model orchestration for multi-step inquiry; building the seven cognitive-scaffold primitives as first-class product surfaces; and closing the Mentor Feedback Loop that measures and improves user cognition over time.

A3.Validation plan and stakeholder outreach (recommended next steps)

For the pre-investment validation phase, we recommend three outreach tracks. First, structured interviews with twenty undergraduate thesis writers, five PhD supervisors, and ten professional autodidacts (journalists, founders, policy analysts) to confirm the cognition-gap framing and willingness-to-pay. Second, two university-level pilots with research-methodology faculty to co-design Guided Modes for undergraduate thesis work. Third, early technical conversations with the Microsoft GraphRAG team and the Anthropic developer-platform team to confirm infrastructure fit and preferred integration surfaces. The IEX Research Xcelerator and the MXT alumni network are natural channels for the academic and founder interviews.

A4.Key references and sources

Microsoft Research's Project GraphRAG overview, including LazyGraphRAG, DRIFT Search, Claimify, and VeriTrail (microsoft.com/en-us/research/project/graphrag).

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